EXPERIMENT NO. 05

DATE OF PERFORMANCE: **GRADE:**

DATE OF ASSESSMENT **SIGNATURE OF LECTURER/ TTA:**

AIM: Implementation of Form controls, Label, Button and TextBox Controls.

VB.NET Form controls:

A Form is used in VB.NET to create a form-based or window-based application.

Using the form, we can build an attractive user interface.

It is like a container for holding different control that allows the user to interact with an application.

The controls are an object in a form such as buttons, Textboxes, Text area, labels, etc. to perform some action.

However, we can add any control to the runtime by creating an instance of it.

A Form uses a System.Windows.Form namespace, and it has a wide family of controls that add both forms and functions in a Window-based user interface.

VB.NET Form Properties

The following are the most important list of properties related to a form. And these properties can be set or read while the application is being executed.

Properties	Description
BackColor	It is used to set the background color for the form.
BackgroundImage	It is used to set the background image of the form.
Cursor	It is used to set the cursor image when it hovers over the form.
AllowDrop	Using the AllowDrop control in a form, it allows whether to drag and drop on the form.
Font	It is used to get or set the font used in a form.
Locked	It determines whether the form is locked or not.
FormBorderStyle	It is used to set or get border style in a form.
Text	It is used to set the title for a form window.
MinimizeBox	MinimizeBox It is used to display the minimum option on the title bar of the form.
IsMDIChild	It is used to authenticate whether a form is a container of a Multiple Document Interface (MDI) child form.
Autoscroll	It allows whether to enable auto-scrolling in a form.
MaximizeBox	It is used to display the maximum option on the title bar of the form.
Language	It is used to specifies the localized language in a form.
AcceptButton	It is used to set the form button if the enter key is pressed.
Top, Left	It is used to set the top-left corner coordinates of the form in pixel.
Name	It is used to define the name of the form.
MinimumSize	It is used to set the minimum height and width of the form.
Enabled	It uses the True or False value to enable mouse or keyboard events in the form.
TopMost	It uses a Boolean value that represents whether you want to put the window form on top of the other form. By default, it is False.

Form Events:

The following are the most important list of events related to a form.

Events	Description
Activated	An activated event is found when the user or program activates the form.
Click	A click event is active when the form is clicked.
Closed	A closed event is found before closing the form.
Closing	It exists when a form is closing.
DoubleClick	DoubleClick The DoubleClick event is activated when a user double clicks on the form.
DragDrop	A DragDrop event is activated when a drag and drop operation is performed.
MouseDown	A MouseDown event is activated when the mouse pointer is on the form, and the mouse button is pressed.

VB.NET Label Control:

In VB.NET, a label control is used to display descriptive text for the form in control.

It does not participate in user input or keyboard or mouse events. Also, we cannot rename labels at runtime.

The labels are defined in the class System.Windows.Forms namespace.

VB.NET Label Properties:

Properties	Description
AutoSize	As the name defines, an AutoSize property of label control is used to set or get a value if it is automatically resized to display all its contents.
Border Style	It is used to set the style of the border in the Windows form.
PreferredWidth	It is used to set or get the preferred width for the Label control.
Font	It is used to get or set the font of the text displayed on a Windows form.
PreferredHeight	It is used to set the height for the Label Control.
TextAlign	It is used to set the alignment of text such as centre, bottom, top, left, or right.
ForeColor	It is used to set the color of the text.
Text	It is used to set the name of a label in the Windows Form.
ContextMenu	It is used to get or sets the shortcut menu associated with the Label control.
DefaultSize	It is used to get the default size of the Label control.
Image	It is used to set the image to a label in Windows Form.
ImageIndex	It is used to set the index value to a label control displayed on the Windows form.

VB.NET Label Events

Events	Description
AutoSizeChanged	An AutoSizeChanged event occurs in the Label control when the value of AutoSize property is changed.
Click	Click event is occurring in the Label Control to perform a click.
DoubleClick	When a user performs a double-clicked in the Label control, the DoubleClick event occurs.
GotFocus	It occurs when the Label Control receives focus on the Window Form.
Leave	The Leave event is found when the input focus leaves the Label Control.
TabIndexChanged	It occurs when the value of Tabindex property is changed in the Label control.
ControlRemoved	When the control is removed from the Control.ControlCollection, a ControlRemoved event, occurs.
TabStopChanged	It occurs when the property of TabStop is changed in the Label Control.
BackColorChanged	A BackColorChanged event occurs in the Label control when the value of the BackColor property is changed.
ControlAdded	When a new control is added to the Control.ControlCollection, a ControlAdded event occurs.
DragDrop	A DragDrop event occurs in the Label control when a drag and drop operation is completed.

VB.NET Button Control:

Button control is used to perform a click event in Windows Forms, and it can be clicked by a mouse or by pressing Enter keys.

It is used to submit all queries of the form by clicking the submit button or transfer control to the next form.

However, we can set the buttons on the form by using drag and drop operation.

VB.NET Button Properties:

Properties	Description
AutoSizeMode	It is used to get or set the auto mode value through which the button can automatically resize in the Windows form.
BackColor	It is used to set the background color of the Button in the Windows form.
BackgroundImage	It is used to set the background image of the button control.
ForeColor	It is used to set or get the foreground color of the button control.
Image	It is used to set or gets the image on the button control that is displayed.
Location	It is used to set the upper-left of the button control's coordinates relative to the upper-left corner in the windows form.
Text	It is used to set the name of the button control in the windows form.
AllowDrop	It is used to set or get a value representing whether the button control can accept data that can be dragged by the user on the form.
TabIndex	It is used to set or get the tab order of the button control within the form.

VB.NET Button Events:

BackColorChanged	A BackColorChaged event is found in button control when the Background property is changed.
BackgroundImageChanged	A BackgoundImageChanged event is found in button control when the value of the BackgoundImage property is changed.
Click	A Click event is found in the button control when the control is clicked.
ContextManuChanged	It is found in button control when the value of the ContextMenu property is changed.
ControlAdded	A ControlAdded event is found in button control when a new control is added to the Control.ControlCollection.
CursorChanged	A CursorChanged event is found in button control when the value of the control is changed.
DoubleClick	When the user makes a double click on the button, a double click event is found in the button control.
TextChanged	It is found in the button control when the value of the text property is changed.
DragDrop	The DragDrop event is found in the button control when the drag and drop operation is completed in the Form.

Furthermore, we can also refer to the VB.NET Microsoft documentation to get a complete list of Button properties and events.

VB.NET Textbox Control:

A TextBox control is used to display, accept the text from the user as an input, or a single line of text on a VB.NET Windows form at runtime. Furthermore, we can add multiple text and scroll bars in textbox control.

However, we can set the text on the textbox that displays on the form.

VB.NET TextBox Properties:

There are following properties of the TextBox control.

Properties	Description
AutoCompleteMode	It is used to get or set a value that indicates how the automatic completion works for the textbox control.
Font	It is used to set the font style of the text displayed on a Windows form.
Lines	It is used to set the number of lines in a TextBox control.
CharacterCasing	It is used to get or set a value representing whether the TextBox control can modify the character's case as they typed.
Multiline	It is used to enter more than one line in a TextBox control, by changing the Multiline property value from False to True.
AcceptsReturn	It is used to get or set a value that indicates whether pressing the enter button in a multiline textbox; it creates a new line of text in control.
PasswordChar	It is used to set the password character that can be a mask in a single line of a TextBox control.
PreferredHeight	It is used to set the preferred height of the textbox control in the window form.
ScrollBars	It is used to display a scrollbar on a multiline textbox by setting a value for a Textbox control.
Text	It is used to get or set the text associated with the textbox control.
Visible	The Visible property sets a value that indicates whether the textbox should be displayed on a Windows Form.
WordWrap	The WordWrap properties validate whether the multiline Textbox control automatically wraps words to the beginning of the next line when necessary.

VB.NET Textbox Events:

Events	Description
Click	When a textbox is clicked, a click event is called in the textbox control.
CausesValidationChanged	It occurs in the TextBox Control when the value of CauseValidation property is changed.
AcceptTabsChanged	It is found in the TextBox control when the property value of the AcceptTab is changed.
BackColorChanged	It is found in the TextBox Control when the property value of the BackColor is changed.
BorderStyleChanged	It is found in the TextBox Control when the value of the BorderStyle is changed.
ControlAdded	It is found when the new control is added to the Control.ControlCollection.
CursorChanged	It is found in TextBox, when the textbox control is removed from the Control.ControlCollection.
FontChanged	It occurs when the property of the Font is changed.
GetFocus	It is found in TextBox control to get the focus.
MouseClick	A MouseClick event occurs when the mouse clicks the control.
MultilineChanged	It is found in a textbox control when the value of multiline changes.

Furthermore, we can also refer to the VB.NET Microsoft documentation to get a complete list of TextBox properties and events.

Program 1: Develop a Windows Form Application using Form Control.

```
Public Class Form1
Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load

End Sub

Private Sub btnGreet_Click(sender As Object, e As EventArgs) Handles
btnGreet.Click
If txtName.Text = "" Then
MessageBox.Show("Please enter your name.")
Else
lblOutput.Text = "Hello, " & txtName.Text & "! Welcome to VB.NET."
End If
End Sub
End Class
```

Output:



Fig 1.1 — Initial state of the form with input field



Fig 1.2 — Output displaying greeting message

Program 2: Write a program to demonstrate the use of Button, Label and TextBox. Take two TextBoxes which takes two inputs from the user (name and number) and shows the message box to print the user details when user presses the Button.

```
Public Class Form1
Private Sub btnShow_Click(sender As Object, e As EventArgs) Handles
btnShow.Click
If txtName.Text = "" Or txtNumber.Text = "" Then
MessageBox.Show("Please fill in both fields.")
Else
MessageBox.Show("Name: " & txtName.Text & vbCrLf & "Number: " & txtNumber.Text)
End If
End Sub
End Class
```

Output:

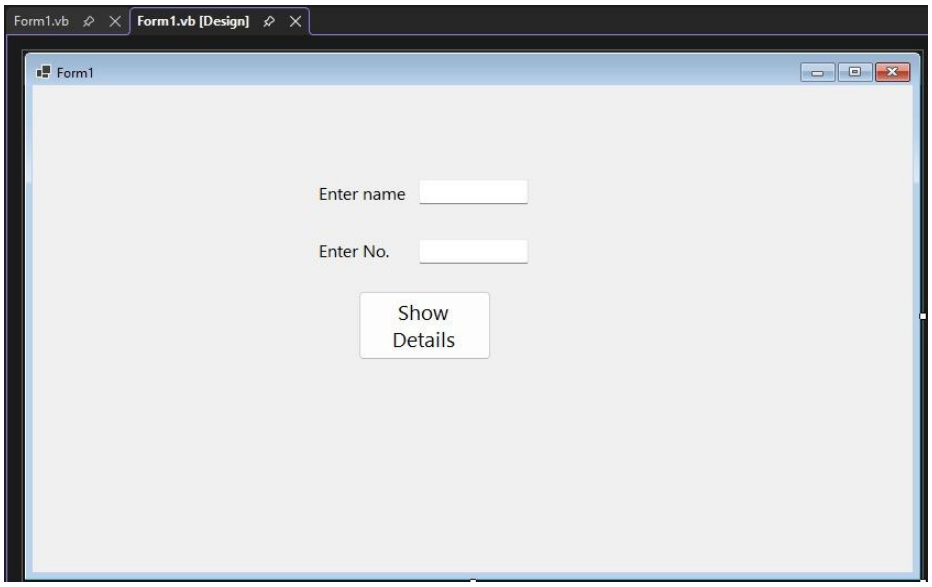


Fig 2.1 — Form design with Label, TextBox and Button controls

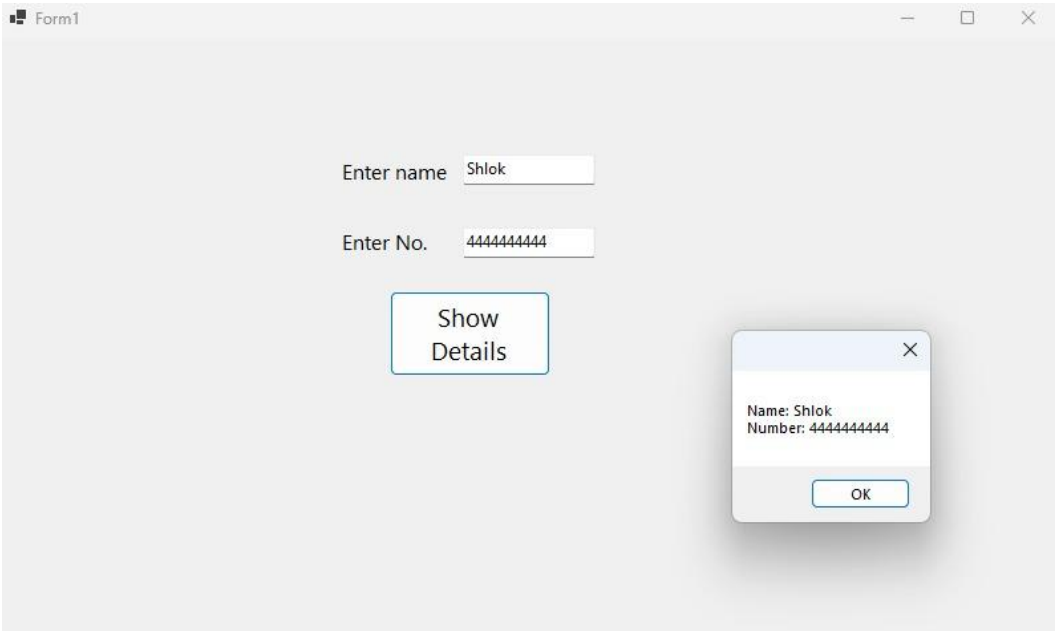
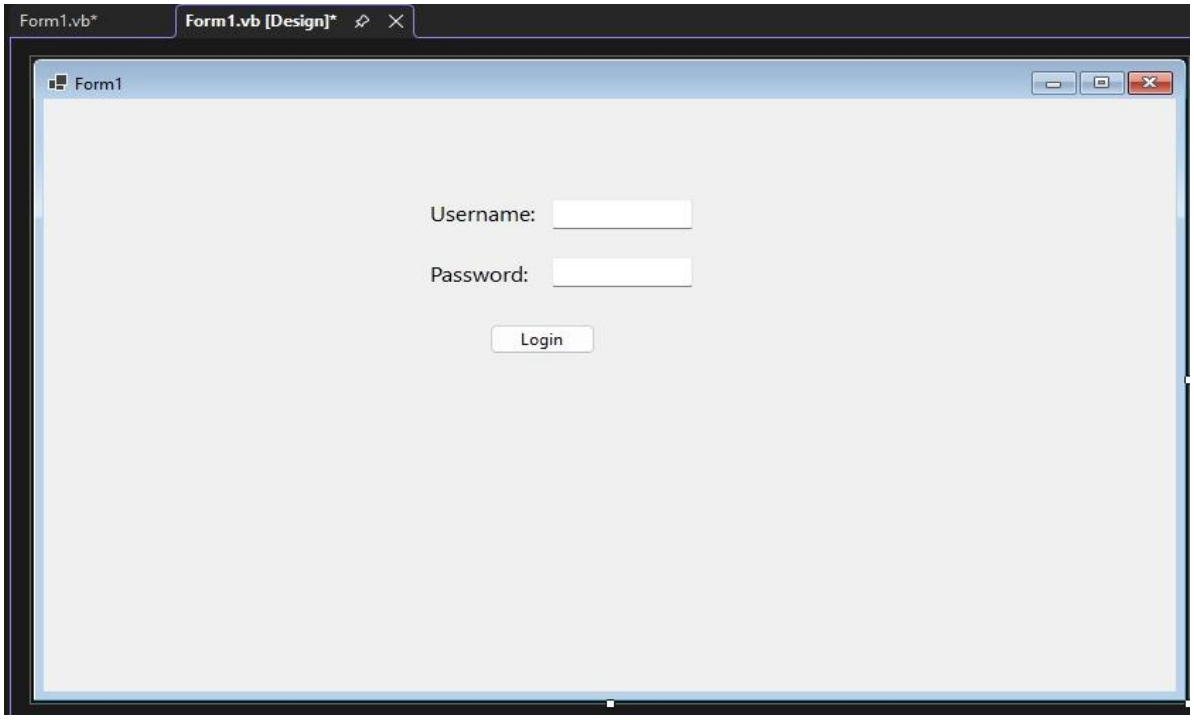


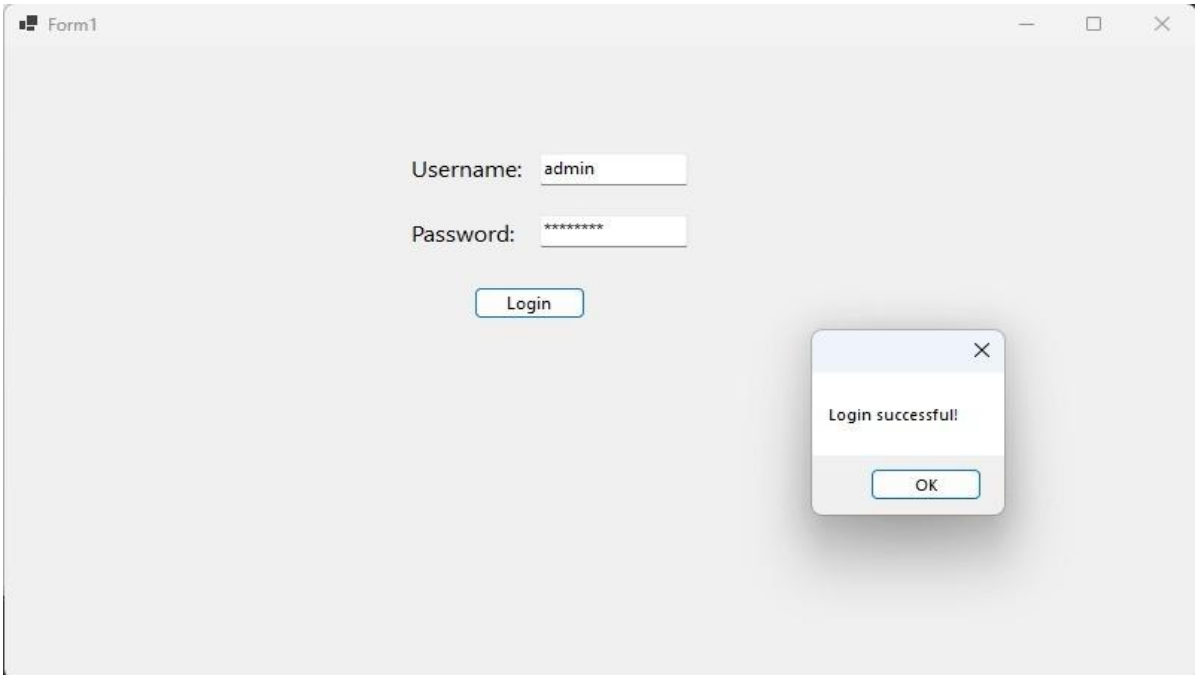
Fig 2.2 — Runtime output showing user details in MessageBox

Program 3: Write a Program on Login form using controls.

```
Public Class Form1
Private Sub btnLogin_Click(sender As Object, e As EventArgs) Handles
btnLogin.Click
If txtUser.Text = "admin" And txtPass.Text = "password" Then
MessageBox.Show("Login successful!")
Else
MessageBox.Show("Invalid username or password.")
End If
End Sub
End Class
```

Output:





Program 4: Design a simple calculator using VB.NET.

```
Public Class Form1

Private Sub btnAdd_Click(sender As Object, e As EventArgs) Handles btnAdd.Click
Dim num1 As Double = CDbl(txtNum1.Text)
Dim num2 As Double = CDbl(txtNum2.Text)
Dim result As Double = num1 + num2
lblResult.Text = "Result: " & result.ToString()
End Sub

Private Sub btnSub_Click(sender As Object, e As EventArgs) Handles btnSub.Click
Dim num1 As Double = CDbl(txtNum1.Text)
Dim num2 As Double = CDbl(txtNum2.Text)
Dim result As Double = num1 - num2
lblResult.Text = "Result: " & result.ToString()
End Sub

Private Sub btnMult_Click(sender As Object, e As EventArgs) Handles
btnMult.Click
Dim num1 As Double = CDbl(txtNum1.Text)
Dim num2 As Double = CDbl(txtNum2.Text)
Dim result As Double = num1 * num2
lblResult.Text = "Result: " & result.ToString()
End Sub

Private Sub btnDiv_Click(sender As Object, e As EventArgs) Handles btnDiv.Click
Dim num1 As Double = CDbl(txtNum1.Text)
Dim num2 As Double = CDbl(txtNum2.Text)
Dim result As Double = num1 / num2
lblResult.Text = "Result: " & result.ToString()
End Sub

Private Sub btnReset_Click(sender As Object, e As EventArgs) Handles
btnReset.Click
txtNum1.Clear()
txtNum2.Clear()
lblResult.Text = "Enter Numbers To see Result"
End Sub
End Class
```

Output:

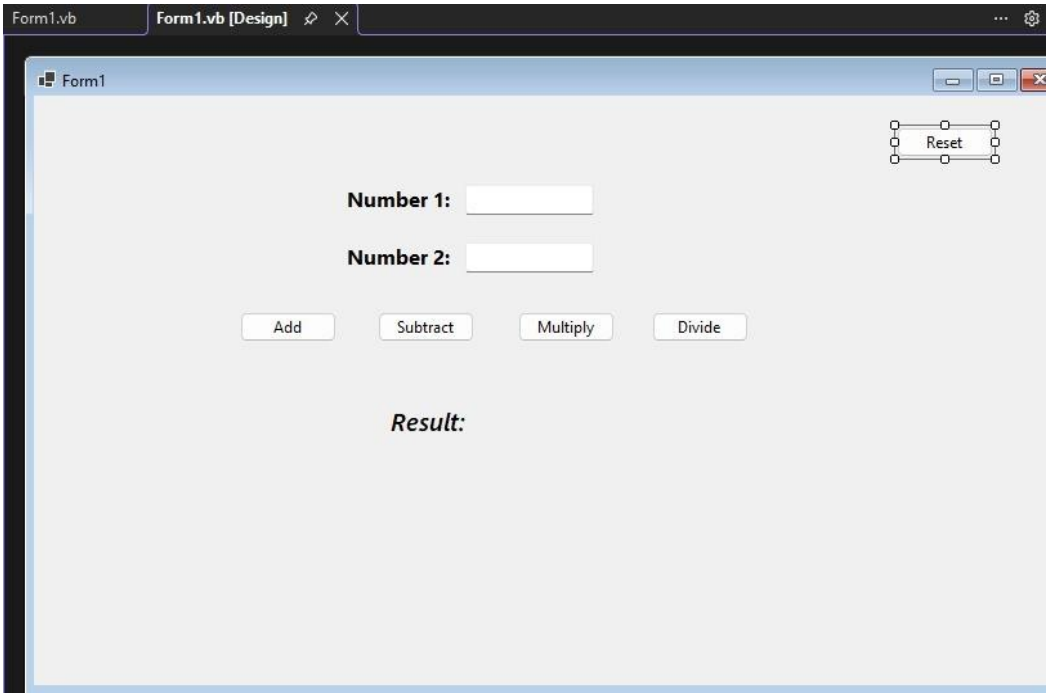


Fig 4.1 — Calculator form design in Visual Studio

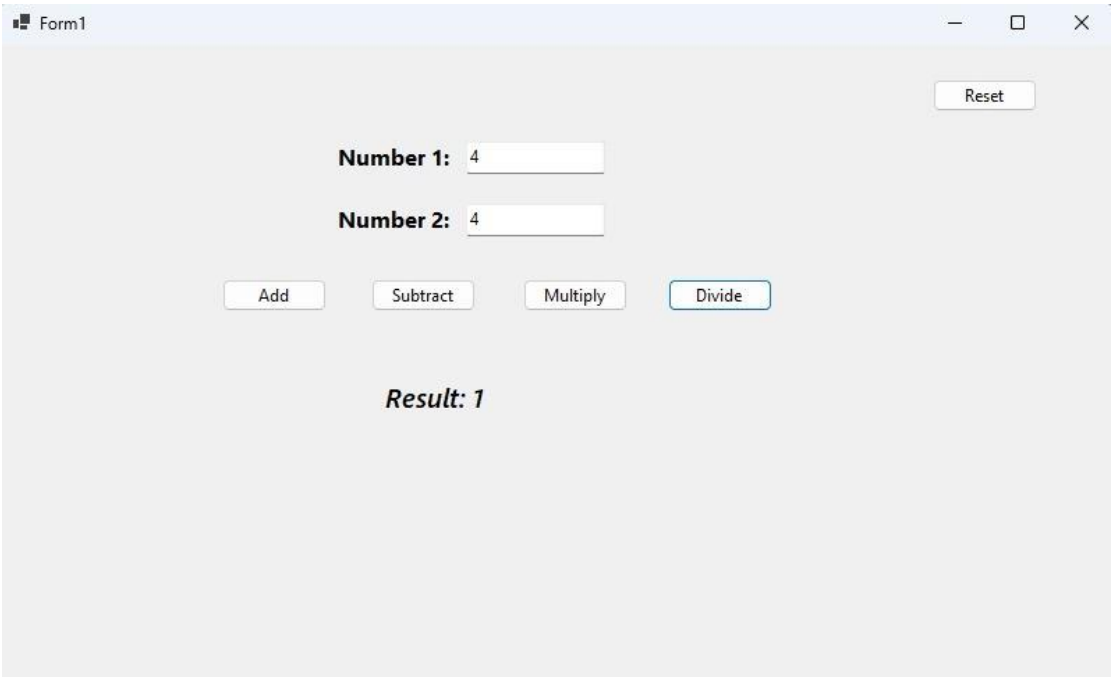


Fig 4.2 — Division operation (4 ÷ 4 = 1)

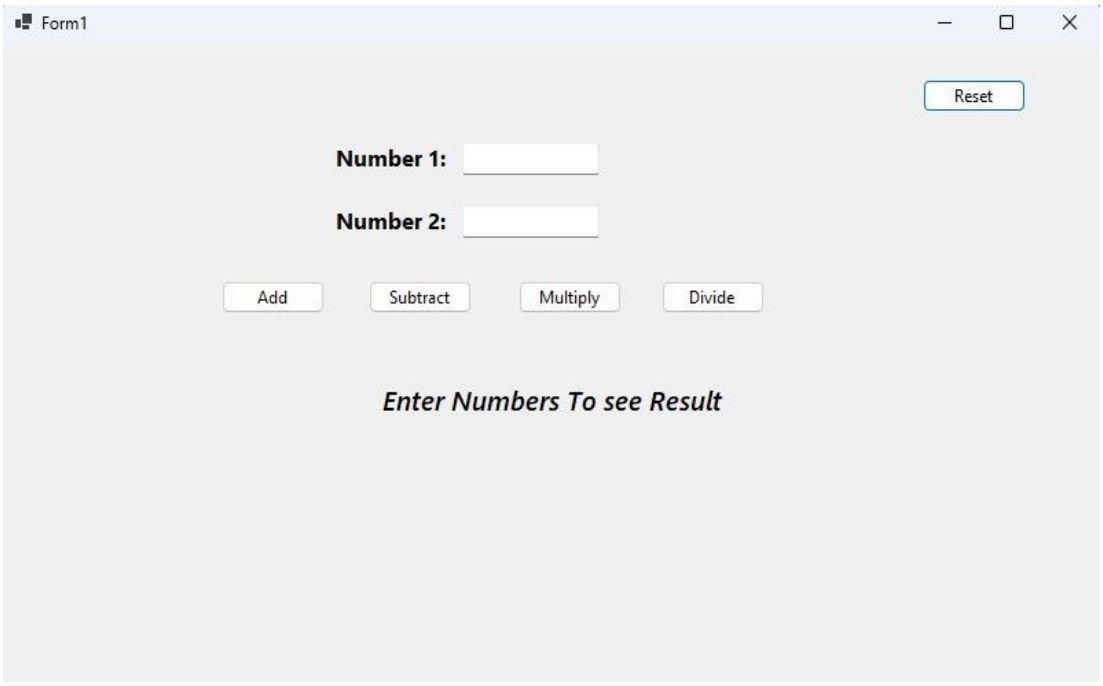


Fig 4.3 — Calculator in reset state

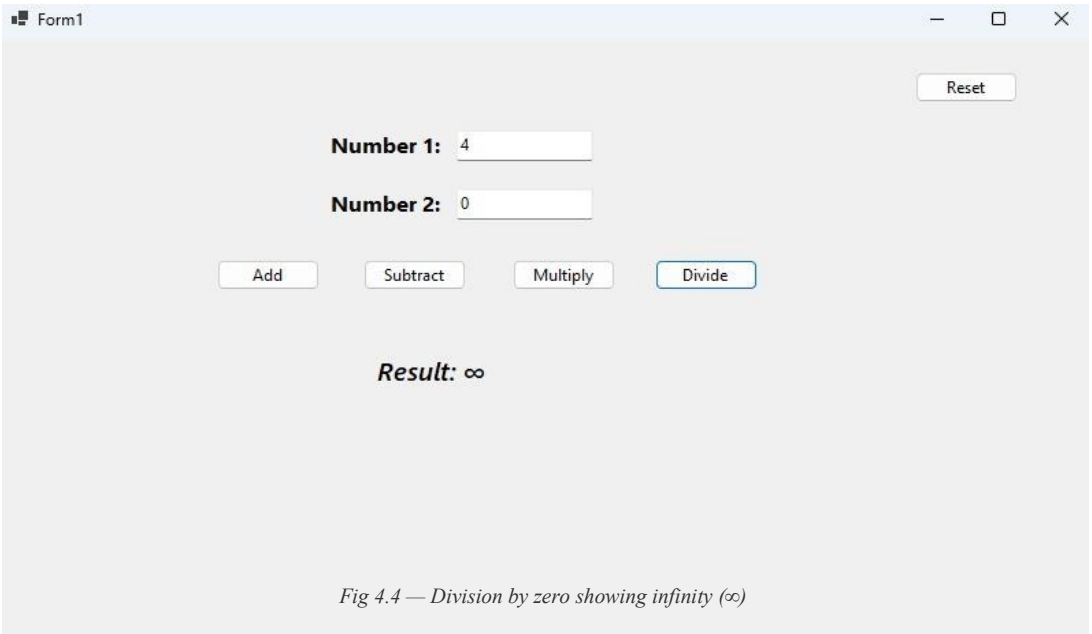


Fig 4.4 — Division by zero showing infinity (∞)